

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: _____

July 4, 2015, Saturday

Department/School: _____

SID: _____

University Physics I
Final Examination

Kuang Yaming Honors School, Nanjing University

Gamma function: $\Gamma(\nu) = \int_0^\infty e^{-t} t^{\nu-1} dt$, $\Gamma(\nu+1) = \nu\Gamma(\nu)$, $\Gamma(1) = \Gamma(2) = 1$, $\Gamma(1/2) = \sqrt{\pi}$.

Gauss integral: $I_n = \int_0^\infty x^n e^{-x^2} dx = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right)$, $I_{2m-1} = \frac{1}{2}(m-1)!$, $I_{2m} = \frac{(2m-1)!!}{2^{m+1}} \sqrt{\pi}$.

Select five out of following six problems.

1.(20pts) An experimenter arranges to trigger two flashbulbs simultaneously, producing a big flash located at the origin of his reference frame and a small flash at $x = 30.0 \text{ km}$. An observer moving at a speed of $0.250c$ in the positive direction of x -axis also views the flashes.

- (a) What is the time interval between them according to the observer?
- (b) Which flash does the observer say occurs first?

2.(20 pts) Galaxy A is reported to be receding from us with a speed of $0.35c$. Galaxy B, located in precisely the opposite direction, is also found to be receding from us at this same speed.

- (a) What recessional speed would an observer on Galaxy A find for our galaxy?
- (b) What recessional speed would an observer on Galaxy A find for Galaxy B?

3.(20pts) Under the standard pressure of atmosphere, 10 kg of water at 20°C is turned into the steam at 250°C . The specific heats at constant pressure for the water and steam are $4187\text{ J/kg}\cdot\text{K}$ and $1670\text{ J/kg}\cdot\text{K}$, respectively. The vaporizing heat for water is $2.25\times 10^6\text{ J/kg}$. Find the change in entropy of the water ΔS .

4.(20pts) Suppose that the molar heat capacity at constant volume for a monatomic ideal gas is $c_v = 3R/2$, where R is the gas constant, and the molar internal energy u of it is not dependent on its volume, i.e., $(\partial u/\partial v)_T = 0$, where v is the molar volume of the gas.

(a) Show that $du = \frac{3}{2}RdT$.

(b) Show that the differential of molar entropy s of the gas is given by $ds = c_p \frac{dT}{T} - Rdp$, where $c_p = \frac{5}{2}R$ is the molar heat capacity at constant pressure of the gas.

(c) Show further that $ds = \frac{3}{2}R \frac{dp}{p} + \frac{5}{2}R \frac{dv}{v}$.

(d) Show that $\kappa_s = \frac{3}{5}\kappa_T$ where $\kappa_s = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_s$ and $\kappa_T = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_T$.

5.(20pts) Consider a gas with a fixed number of particles. The fundamental equation of thermodynamics of it is given by $dU = TdS - pdV$.

(a) Show that $dF = -SdT - pdV$ where $F = U - TS$ is the Helmholtz free energy of the system, and that $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$.

(b) Show that $dS = \frac{C_V}{T}dT + \left(\frac{\partial p}{\partial T}\right)_V dV$, where C_V is the heat capacity of the gas at constant volume.

(c) Show that $dU = C_V dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV$.

(d) Show that Joule coefficient $K = \left(\frac{\partial T}{\partial V}\right)_U \neq 0$ for the van der Waals gas with one mole molecules.

(Hint: The van der Waals equation for one mole of molecules can be written as $p = \frac{RT}{V-b} - \frac{a}{V^2}$)

6.(20pts) The ideal gas of total number of molecules N obeys the Maxwell velocity distribution

$$dn = C \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z$$

where m is the mass of the molecule, T the temperature, and k_B the Boltzmann constant. And the multiplier C is to be determined.

(a) Determine the multiplier C in terms of m, k_B, T and N .

(b) Find $\langle v_x \rangle$ and $\langle v_y \rangle$, the averages of the velocity components v_x and v_y .

(c) Find $\langle v_x^2 \rangle$, $\langle v_y^2 \rangle$ and $\langle v_x v_y \rangle$.

(d) Find the fluctuation of the x -component of velocity $\Delta v_x \equiv \sqrt{\langle (v_x - \langle v_x \rangle)^2 \rangle}$.